BillDrop - 1-Page Plan

Vizdale Full-Stack Developer Internship | Milestone 1 | Name: _____

1. Client and problem

Client (made-up): Farook, Manager, Irumbu Mart, a hardware wholesaler in Salem with 45 suppliers and about 150 paper bills a week. The accountant spends 2 full days typing bills into Excel. Last month one supplier's Rs 64,000 bill was paid twice.
How might we turn a week of paper supplier bills into clean, checked records in minutes instead of 2 days, and catch errors before money is paid?

2. Users

Role	Can	Cannot
Accountant	Log in, upload bills, check and correct, save, download Excel, draft a message to a supplier	Skip the review: nothing is saved until checked
Manager	Log in, see supplier totals and bills with problems	Upload, correct or save a bill

3. The one flow for M2

Accountant logs in, uploads one PDF/JPG/PNG bill, OCR reads the text, AI turns it into fields (supplier, bill number, date, lines, total), the review screen shows the bill beside the fields, the accountant corrects and presses Save, and the record is stored in the database.

4. Tools and hosting (all free)

Layer	Choice	Why
Frontend	React (Vite), hosted on Vercel	Fast to build, free public link
Backend	Java 17 + Spring Boot, hosted on Render	Server-side role checks (401/403)
Database	MySQL (free cloud MySQL)	Tables from the brief: people, suppliers, batches, bills, bill_lines
OCR / AI	Free OCR service + free AI API, with saved sample answers for tests	No outside calls in tests
Checks	GitHub Actions: build + secret scan	Green tick on every push

5. Plan by milestone

Milestone	What will be done
M1 (7 Nov)	GitHub project, first React page online, automatic check passing, this plan
M2 (13 Nov)	Tables + seed script, login for both roles, upload one bill, OCR, AI reading, review screen and save, 60-second recording
M3 (21 Nov)	20 bills at once with background jobs, live cards, wrong-total and duplicate checks, supplier totals, Excel export, AI draft message, 10 tests, 10 real testers
M4 (28 Nov)	Bug fixes, README, 2-minute video, AI notes, speed score 85+, progress template

6. Risks and how I will handle them

Risk	Plan
Free OCR limits file size (1 MB, 3 pages)	Shrink files before sending; use small made-up bills
AI answer in wrong format	Check the format; if wrong, send bill to review with empty fields, never crash
Free hosting sleeps or is slow	Open the link before demos; keep the page light
Secret keys leaking	Keys only in environment variables; secret scan in the automatic check
Short on time	Finish M2 flow first; skip the stretch goal

7. Not building

GST rules or tax advice, paying bills, multi-page or handwritten bills, supplier management screens, polished design.